

BrickAtlas LDraw-1: Instance-Grounded Visual and Structural Reasoning over Original Brick Assemblies

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Abstract

Reasoning about brick assemblies requires more than recognizing an object’s silhouette: an evaluator must identify the exact part, preserve its geometry, and distinguish what is visible from what is structurally known. We introduce BrickAtlas LDraw-1, a benchmark of 24 original LDraw/OMR assemblies, 15,334 part instances, and 617 questions across eleven task families. A stable instance number links rendered observations, source records, connector graphs, and executable scene edits. The task engine separates visual perception, source-data reasoning, graph reasoning, and instruction-following edits, with explicit evidence and success contracts. We provide real 3D inputs, functional task decompositions, source-level statistics, and a reproducible evaluation protocol. Independent reparsing verifies every retained source instance; all reference solutions and 75 public-action solver outputs satisfy their contracts. Answer-prior analysis exposes uneven task coverage and elementary evidence controls. These results validate the software interface, not learned-model ability or physical stability. Model, human, robustness, and physical-validation experiments are specified with unfilled result tables. The release makes assembly reasoning inspectable while retaining unresolved structural evidence.

1. Introduction

An assembly assistant must connect language to individual physical parts. “Remove the support” is ambiguous when several pieces have the same shape and color. A convincing rendering can hide disconnected geometry. A connector graph can support an exact adjacency answer while remaining insufficient to predict whether an object will stand under gravity. Evaluating all three cases with a single notion of assembly success obscures the source of an error.

Recent work has advanced interactive reconstruction [13], manual interpretation [14], and graph-backed brick generation [7]. These settings motivate a complementary question: can a model ground a request in a fixed, authentic assembly and reason correctly under the evidence

provided? This question is useful even before introducing robotics. An assistant that selects the wrong instance or treats missing connector data as proof of a floating part cannot provide reliable downstream instructions.

We introduce **BrickAtlas LDraw-1**. Its starting point is an unchanged original source assembly, not a generated arrangement that resembles one. Each occurrence of a part receives a stable B0001-style identifier. Questions, numbered renders, public evidence, action execution, and review records share that identity. Repeated part types remain separate instances. The source montage in Fig. 1 shows vehicles, creatures and larger structures; all are rendered from retained LDraw geometry.

Our design separates four observation contracts. Visual tasks identify color or corresponding part types from numbered images. Source-data tasks use supplied coordinates, coverage flags or author records. Graph tasks operate on recognized connector pairs. Scene-edit tasks restore an instance’s visibility or follow a source-step window at fixed poses. This separation makes exact evaluation possible without treating an editor operation as a mechanically feasible insertion.

The release has three contributions. First, it provides a source-preserving corpus with instance-level grounding and attributable dependencies. Second, it defines eleven inspectable task families, including set-valued graph answers and executable action sequences. Third, it provides a reproducible evaluation specification, descriptive analyses and failure-review artifacts that distinguish measured evidence from experiments still to be conducted. The current manuscript is a benchmark development draft: it reports no learned-model leaderboard and no claim of certified physical stability.

2. Related Work

Brick representation and assembly. Break and Make [13] evaluates interactive structural understanding through disassembly and reconstruction. Translating a Visual LEGO Manual [14] connects visual instructions to executable plans. BrickNet [7] represents typed interfaces and generates assemblies with connector graphs; we use



its public connector implementation to audit fixed source placements. BrickGPT [10] targets text-conditioned stable, buildable structures. LDraw-1 evaluates questions over existing sources and does not reproduce those methods’ generative or robotic results.

Compositional visual reasoning. CLEVR [6] and GQA [5] make question semantics and compositional operations explicit. We adopt this principle for numbered CAD instances, but do not inherit their balancing procedures or validation results. BLINK [2] emphasizes basic perceptual failures, while LEGO-Puzzles [11] studies multi-step spatial reasoning. A source part number in our reference solution is a CAD identity, not a guarantee that all visually equivalent parts share one label.

Physical reasoning and evaluation. PhyBlock [9] evaluates physical understanding and planning under its simulation contract. Our graph and scene operations have narrower semantics: recognizing a hinge does not establish its load capacity or simulated dynamics. Dataset documentation principles [3] motivate explicit provenance, coverage and intended use. Shortcut learning [4] and option-order sensitivity [15] motivate the answer-prior and permutation controls specified below.

3. Benchmark Representation

3.1. Source Identity and Numbered Observations

An assembly is a collection

$$M = \{(b_i, p_i, c_i, T_i, h_i)\}_{i=1}^n, \quad (1)$$

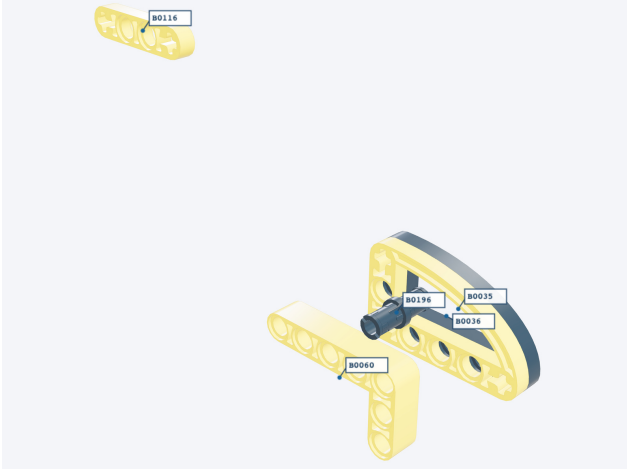
Table 1. Task positioning. Success criteria depend on the observation and action contract; no numerical cross-benchmark ranking is implied.

Setting	Evaluation object
Break and Make	Interactive structural reconstruction
Visual LEGO Manual	Visual instructions to executable plans
BrickNet / BrickGPT	Generated brick assemblies
LEGO-Puzzles	Multi-step spatial reasoning
PhyBlock	Physical understanding and planning
LDraw-1	Fixed sources, numbered instances, explicit evidence and scene edits

where b_i identifies an instance, p_i is its LDraw part type, c_i its source color, T_i its source transform, and h_i its hierarchy path. Nested MPD submodels are expanded with source file and line provenance. Dependency closure and geometry are preserved. A documented axis and unit conversion maps LDraw coordinates to the viewer’s millimeter frame [8, 12].

Let H be the original file’s SHA256 digest. The number $\ell_i = \mathbb{B} \parallel \text{pad}_4(i)$ maps to one instance within H ; (H, b_i) is the complete identity. Numbering does not identify a type or describe a role. Thus two identical pins have different references even when a language description would be identical. Fresh source parsing verifies the stored type, color, transform and hierarchy for every instance.

Numbered leader lines connect the relevant IDs to projected geometry. Fig. 2 shows an operand-isolation view from set 42004. At most twelve labels appear simultaneously; each operand remains individually accessible through inventory lookup and isolation. Occlusion is not solved by a label alone. The interactive contract therefore



permits rotation, zoom and isolation, while the static contract uses a frozen numbered input. All 140 visual questions have an exported operand image. Their public records omit source attributes and answers.

3.2. Typed Structural Evidence

The connector audit produces a multiset R of records $(b_i, b_j, k_i, k_j, f, \theta)$, where k_i, k_j index connectors, f is an interface family and θ contains matching parameters. Brick-Net 0.1.0 recognizes stud, fixed, axle, hinge and ball families. For graph questions we use the simple undirected projection

$$E = \{\{b_i, b_j\} : \exists r \in R \text{ linking } b_i, b_j\}. \quad (2)$$

Repeated connector records do not increase the degree. Unsupported definitions remain explicit. Definition coverage measures catalog support, while connected components describe the recognized graph; neither is a complete physical connectivity score.

Connector inference uses the upstream alias rules and tolerances. Its parser normalizes rotation matrices, so we record the maximum deviation for each source. Rendering retains the original transforms, and inset-mesh queries use the original alias-transformed poses. Intersections are candidate evidence: friction-fit and gear geometry can intersect an approximation without constituting a defect. The release retains 573 pairs without recognized mating interfaces for further adjudication. No source parts are moved to improve these counts.

3.3. Executable Task Contract

A task $q = (H, \mathcal{B}, o, f, x, y^*)$ binds source H , numbered operands $\mathcal{B}$, observation contract o , family f , public input x and reference output y^* . Model-facing records contain the first five fields, with answers stored separately for review.

Visual, source-data, graph and scene-edit inputs are denoted V, S, G and E , respectively.

For scene edits, a state contains facts F and a visible instance set V . An action a specifies required facts R_a , forbidden facts B_a , additions A_a , deletions D_a and cost c_a . It is legal if $R_a \subseteq F$, $B_a \cap F = \emptyset$, and its cumulative cost fits the task budget. Its fact transition is

$$F' = (F \setminus D_a) \cup A_a. \quad (3)$$

The corresponding display transition changes visibility at original poses. Unknown actions, repeated completed steps, missing prerequisites and budget overruns are rejected. A rejected transition leaves the display state unchanged. Goal and prohibited facts define success.

4. Task Construction

4.1. Functional Families and Applicability

Table 2 decomposes every family into its operations. The four organizational layers are atomic, metacognitive, procedural and integrative. They contain 213, 201, 130 and 73 tasks, respectively. These labels describe the intended operation, not an experimentally calibrated hierarchy of difficulty. In particular, some metacognitive items are elementary checks of an explicit evidence statement.

The generator samples instances deterministically from each source, with a size-dependent cap of two to five targets. It creates tasks only when the evidence supports the contract. Type correspondence requires another instance of the same source type and different-type distractors. Distance questions use geometry bounding-box centers, not mass centers, and reject near ties. Connector classification supplies the record’s family explicitly; its purpose is evidence reading, not visual contact inference.

Neighbor questions require the exact set of adjacent candidates. Graph-deletion questions remove a vertex and incident edges from the supplied local induced graph, then count its connected components. Disconnected distractors can contribute to that count. This operation does not imply that the selected brick can physically be extracted without moving others. Coverage questions identify unsupported definitions, and evidence-limit questions ask whether the supplied audit establishes gravitational stability. The latter always has the semantic answer “not established by this evidence.”

There are 469 single-choice, 73 multiple-choice and 75 action tasks. Options are deterministically shuffled by task identity. This removes a fixed construction order but does not balance semantic answers. The applicability rules produce unequal counts across families and sources; we retain those differences rather than manufacture tasks without their prerequisites.

Table 2. Complete task decomposition. Layers: A, atomic; M, metacognitive; P, procedural; I, integrative. Inputs: V, visual; S, source data; G, connector graph; E, scene edit. Operations describe the actual task generator rather than a learned model’s hidden reasoning.

Family	N	Layer	Input	Functional decomposition
Color	73	A	V	Bind instance; inspect color; select option.
Part type	67	A	V	Bind target and candidates; ignore pose/color; match source type.
Distance	73	A	S	Read bounding-box centers; compute Euclidean distances; select minimum.
Interface	87	M	G	Read supplied connector record; map its family to an option.
Neighbors	73	M	G	Bind target; collect one-hop neighbors; return exact candidate set.
Coverage	17	M	S	Read definition-coverage flags; identify the unsupported instance.
Evidence limit	24	M	S	Inspect evidence scope; distinguish CAD evidence from stability proof.
Step lookup	55	P	S	Bind instance; filter author records; return first expanded step.
Step sequence	17	P	E	Read source window; satisfy predecessor facts; reveal steps in order.
Restoration	58	P	E	Bind missing ID; check missing fact; restore original visibility.
Graph deletion	73	I	G	Induce local graph; delete target vertex and edges; count components.

Table 3. Set 42102 source-step trace. Each action also forbids its own completed fact and costs one unit; the final goal is done-6.

Step	Prerequisite	Added IDs	$ V $	New fact
1	start	B0001-B0003	3	done-1
2	done-1	B0004-B0005	5	done-2
3	done-2	B0006-B0008	8	done-3
4	done-3	B0009-B0010	10	done-4
5	done-4	B0011-B0012	12	done-5
6	done-5	B0013-B0014	14	done-6

Table 4. Measured scale bands. D1: ≤ 150 instances; D2: 151–400; D3: 401–1,000; D4: $> 1,000$. Bands measure source size only.

Band	Sources	Instances	Tasks
D1	5	501	83
D2	9	2172	175
D3	2	1440	59
D4	8	11221	300

4.2. Six-Step Worked Example

Seventeen sources contain usable author STEP provenance. Expanded author steps preserve source hierarchy and ordering; editorial groups or file-list positions never substitute for instructions. The source-sequence family takes the first usable window of up to six steps and encodes predecessor facts. It evaluates following the supplied author order, not discovering a new assembly plan.

Fig. 3 shows all six states for set 42102 in reading order. Table 3 lists the newly revealed IDs and state facts. Step lookup asks where an instance first appears; step sequence asks for the legal ordered program. Color, type and distance families bind numbered parts in the source, while graph deletion operates on supplied connector evidence. These are different questions about a common assembly, not six unmeasured levels of physical difficulty.

Restoration instead hides a target in a display copy and accepts only its restoration. For target B0094, the initial, accepted and rejected scenes contain 128, 129 and 128 visible instances. The supplement shows these states. Neither restoration nor STEP replay interpolates an insertion path; no collision-free or force-feasible trajectory is implied.

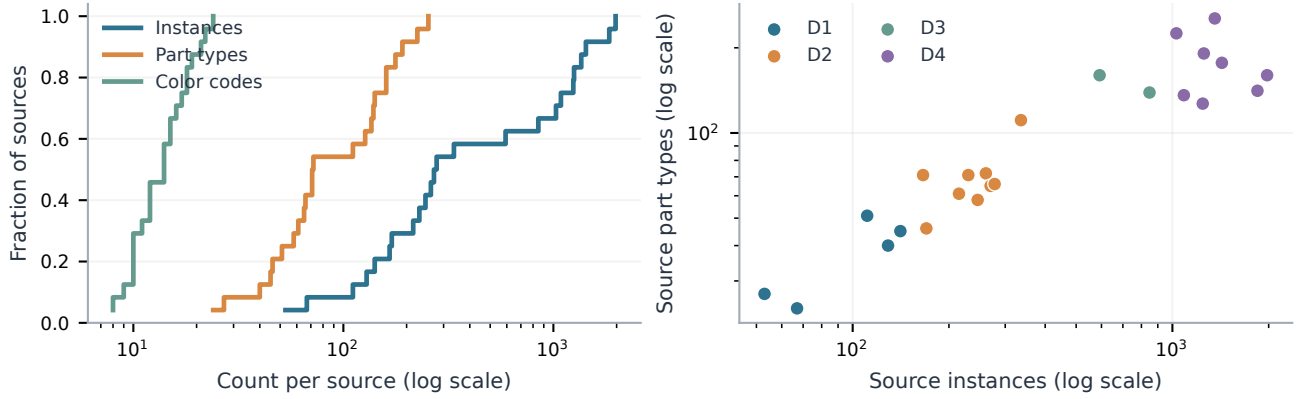
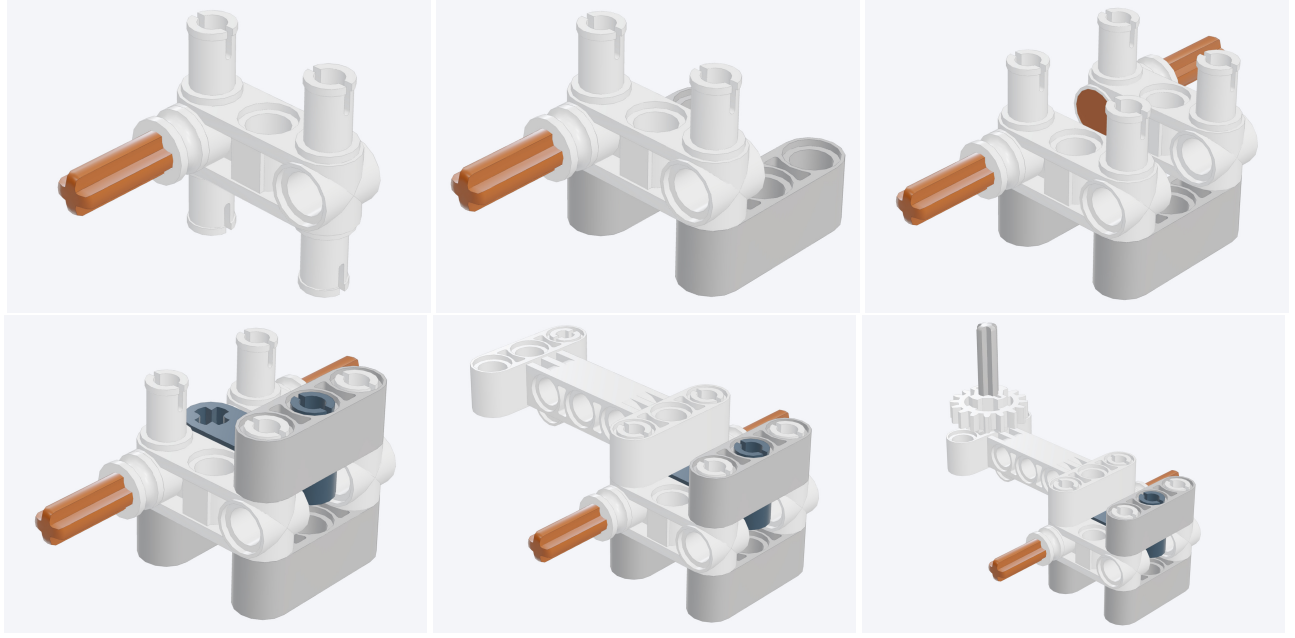
5. Corpus and Descriptive Analysis

5.1. Acquisition, Scale and Diversity

The preparation inventory contains 43 candidates, of which 29 complete sources were converted. Fourteen were excluded for recorded preparation issues such as unresolved geometry, attribution gaps, size limits, singular transforms or loader identity mismatch. Five further sources were excluded for duplicate placements or definition coverage below 90%. This threshold limits unsupported graph evidence; it is not a physical acceptance probability. The remaining 24 sources represent 24 set families. Unknown parts remain in their original scenes.

Table 4 gives the scale bands. The corpus contains 1,002 distinct source part-type strings and 49 color codes; these are representation vocabularies, not canonical mechanical equivalence classes. Source sizes range from 53 to 1,978 instances, with median 274. Fig. 4 plots the empirical distributions and the relation between instance count and type count. Larger scenes do not simply repeat one small part inventory.

We retain one primary build per set family. Geometry fingerprints remove labels, colors, instance ordering and global translation; all retained fingerprints differ. The 276



source pairs also receive an inventory screen,

$$J(A, B) = \frac{\sum_p \min(c_A(p), c_B(p))}{\sum_p \max(c_A(p), c_B(p))}, \quad (4)$$

where $c_A(p)$ counts occurrences of type p . The largest observed Jaccard value is 0.450, between the large tractor and jet. A centered, scale-normalized nearest-center score supplies an additional shape screen. Both are descriptive: shared inventories and orientation sensitivity prevent interpreting them as proof of independence.

5.2. Coverage and Structural Heterogeneity

Fig. 5 reports all family-by-band counts. D3 has only two sources, whereas D4 has eight; source-order families additionally depend on author metadata. Consequently, a pooled

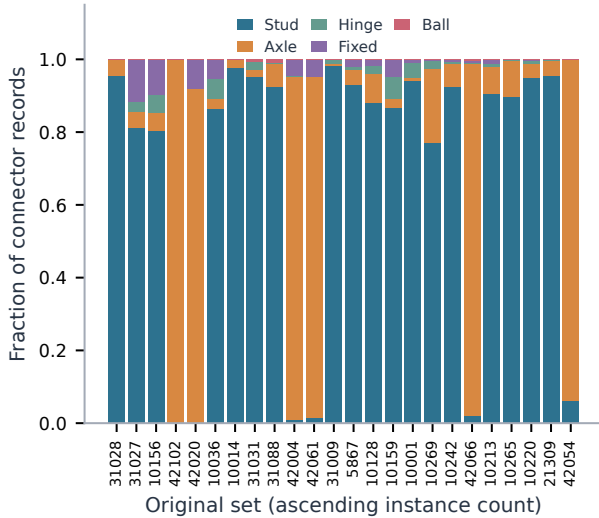
score can change with source mix even if within-family performance is unchanged. Per-family and source-macro reporting are necessary.

Fig. 6 gives within-source connector composition. The audit records 34,877 matches: 27,132 stud, 6,976 axle, 411 hinge, 326 fixed and 32 ball records. Stud interfaces dominate globally, while Technic sources contribute more axle and joint evidence. Records can repeat a part pair; these counts are not independent contacts and do not measure mechanical strength.

5.3. Answer Priors and Elementary Controls

Fig. 7 compares single-choice priors by family. Across 469 items, always choosing A yields 26.652% accuracy; uniform choice has expected accuracy 27.342%, accounting for each item's option count. These values should not be gen-

Color	10	18	7	38
Part type	7	16	7	37
Distance	10	18	7	38
Interface	13	33	9	32
Neighbors	10	18	7	38
Coverage	2	6	2	7
Evidence limit	5	9	2	8
Step lookup	4	14	4	33
Step sequence	2	7	1	7
Restoration	10	18	6	24
Graph deletion	10	18	7	38
	D1	D2	D3	D4
	Source scale band			



eralized to exact-set or action outputs. The evidence-limit family has a constant semantic answer, so high accuracy there alone provides little evidence of reasoning. Graph-deletion answers are also concentrated. Per-family performance must therefore accompany any aggregate.

Semantic concentration is computed in-sample for shared answer vocabularies; instance-valued answers have no cross-source semantic majority. It is a diagnostic, not a held-out baseline. No balancing or adversarial filtering result is claimed. Future balancing must be versioned so that the present questions and measured counts remain reproducible.

Table 5. Measured software and choice controls. Reference checks use privileged answers; the action solver uses public preconditions. Rates are contract validation or single-choice controls, not MLLM scores.

Control	N	Rate (%)
Reference solutions	617	100.0
Public-precondition solver	75	100.0
Invalid controls rejected	1606	100.0
Always A (single choice)	469	26.7
Uniform choice (expected)	469	27.3

6. Evaluation Protocol

6.1. Conditions, Metrics and Reproducibility

The proposed primary evaluation uses each task’s designated public input: V (140 items), S (169), G (233), and E (75). Static and interactive visual results must be reported separately. An image-only system should not be scored as if it received the graph evidence, and a text-only control should not receive answer-bearing visual source attributes. Every run must freeze its checkpoint or API revision, prompt, decoding settings, image resolution, tool budget, timeout and task manifest.

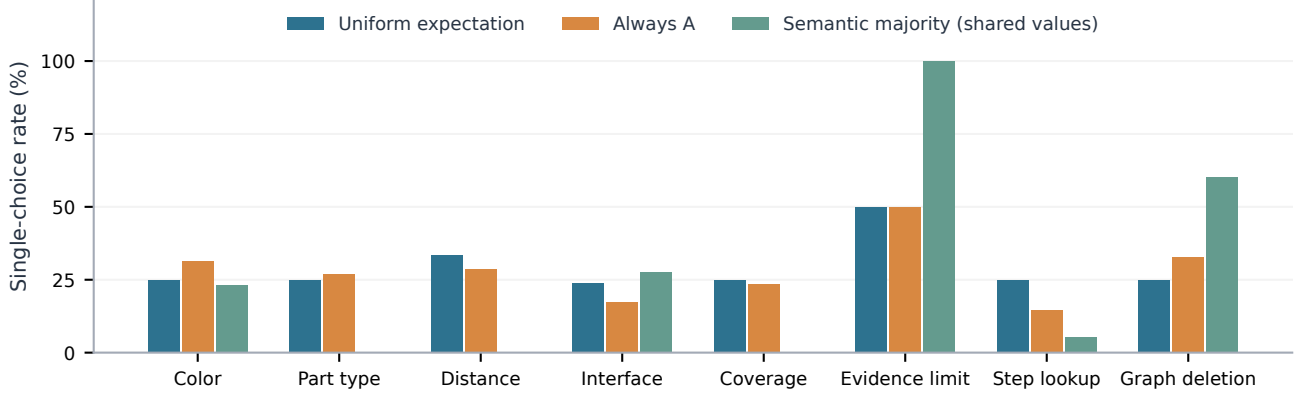
Single-choice success is exact option identity; multiple-choice success requires the exact set, with duplicate IDs invalid. Action success requires a legal complete execution and the specified goal state. Formatting errors, refusals and timeouts remain failures in the denominator and receive separate reason codes. Let $s_q \in \{0, 1\}$ be task success and Q_m the eligible questions for source m :

$$\text{Micro} = \frac{\sum_q s_q}{|Q|}, \quad (5)$$

$$\text{Macro} = \frac{1}{|\mathcal{M}_Q|} \sum_{m \in \mathcal{M}_Q} \frac{\sum_{q \in Q_m} s_q}{|Q_m|}. \quad (6)$$

For a restricted modality or family, $\mathcal{M}_Q$ contains only sources with eligible questions. Report exact success, valid-output rate and action-goal success separately. A partially correct neighbor set may receive secondary set F1, but does not count as exact success.

Tasks within a source are correlated. Proposed confidence intervals resample source groups rather than individual questions, using a fixed seed and 10,000 bootstrap replicates [1]. For trained methods, folds must group all instances, views, questions and alternative builds by source set; neighboring source inventories require additional leakage inspection. No train/test split or confidence interval is presented as already established.



6.2. Measured Validation

Independent parsing reproduces all 15,334 source instances. Checks cover identities, types, colors, transforms, hierarchy, dependencies and mesh hashes. All 617 reference answers pass their contracts, all 75 action tasks are solvable from public preconditions, and 1,606 invalid controls are rejected (Table 5). These checks establish consistency between construction and execution. They do not establish perceptual grounding, task difficulty or physical truth.

Every source loads its complete instance count in the viewer. Captures include four canonical views, numbered operand images and worked action frames. Interaction checks exercise isolation, explosion, source replay, rejected operations and review persistence. Review records bind the source hash, task ID, numbered operands, decision, reason and timestamp. A failed review requires a reason; software validation never writes a human approval.

6.3. Planned Model Comparisons and Ablations

Table 6 specifies candidate model families for the main comparison. Exact checkpoints and service revisions remain to be frozen. All numeric result cells are intentionally empty because no learned model has been evaluated on this release. The supplement provides per-family, scale, action and resource tables with the same missing-result convention.

Table 7 isolates access to images, view control, numbered isolation and graph evidence. Withholding a required graph is an information ablation, not the same task condition. The public action solver row in this planned table concerns standardized resource logging; its existing contract success is reported separately above. Tool-using agents must log observations and calls, including failed calls, rather than only their final answers.

Robustness tests permute IDs consistently across operands, overlays, evidence and scoring, or permute option order while preserving semantic answers. Camera changes test visual sensitivity. Color changes apply only to

Table 6. Proposed learned-model comparison: exact success (%) on designated inputs. V/S/G/E are modality scores; Macro averages sources. Candidate families are planning entries, not completed runs. All blank result cells mean *not run*.

Candidate family	V	S	G	E	Micro	Macro
SmolVLM						
Qwen3-VL						
Llama 4						
GPT-4.1						
Gemini						
Claude						

Table 7. Proposed information and tool ablations. Acc. and Valid are percentages; Calls is mean tool calls per task. Each comparison uses a frozen model and its eligible subset. Cells are unrun.

Condition	N	Acc.	Valid	Calls
Static numbered image	140			
+ rotate / zoom	140			
+ ID isolation	140			
Text without image	140			
Structured public input	477			
Graph edges withheld	233			
Public action solver	75			

type-correspondence questions that explicitly ignore color, and remain evaluation copies of fixed sources. Equivalent edge-order changes test serialization sensitivity. Paired consistency measures agreement after mapping outputs back to canonical identities; correct agreement and consistently wrong answers must be distinguished.

Human review should separately assess ID legibility, visual answer ambiguity, source attribution, graph semantics and intersection candidates. Independent dual review, agreement and adjudication counts are proposed, with no human-pass rate yet available. Physical validation requires addi-

tional assumptions about materials, contact and loads; its results must remain separate from question accuracy.

7. Discussion and Limitations

LDraw-1 exposes a useful distinction between object authenticity, evidence completeness and answer correctness. Preserving a real source prevents arbitrary geometric edits from silently changing the question, but does not prove that contributor-authored CAD is physically valid. The 573 unresolved intersection candidates, unsupported definitions and inference tolerances remain material limitations. Source scenes can also contain several legitimate independent objects.

The corpus is small and uneven. Several families are evidence-reading controls, and the public-action solver’s perfect contract success shows that these action tasks do not demand discovering mechanical plans. Original part-type strings can distinguish visually near-identical variants; ambiguous correspondence items require human adjudication. D1–D4 encode size, not measured cognitive difficulty.

Commercial sets may be familiar from pretraining. Public review artifacts include answers, so the development website is not a hidden test server. Evaluation must restrict models to designated inputs and audit tool access. Earlier synthetic-release model pilots and physical claims are excluded from this release.

8. Conclusion

BrickAtlas LDraw-1 provides a common identity and evidence interface for visual, structural and procedural questions over original assemblies. Its current contribution is a reproducible corpus and executable evaluation specification, supported by measured source statistics and software controls. The unfilled model and validation tables identify the empirical work needed before claiming model rankings, calibrated difficulty or physical reliability.

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